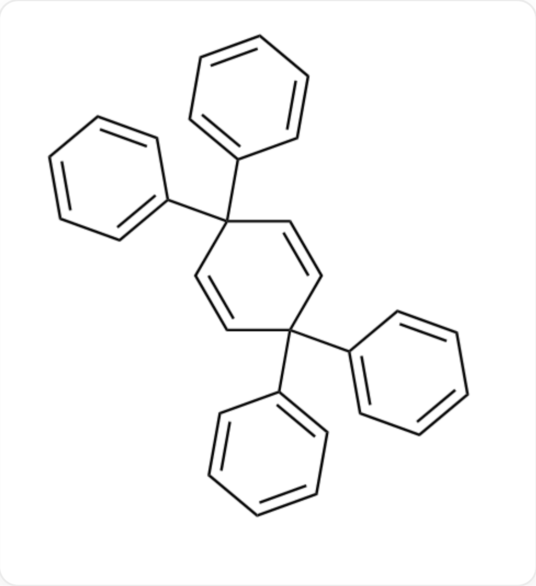
Question

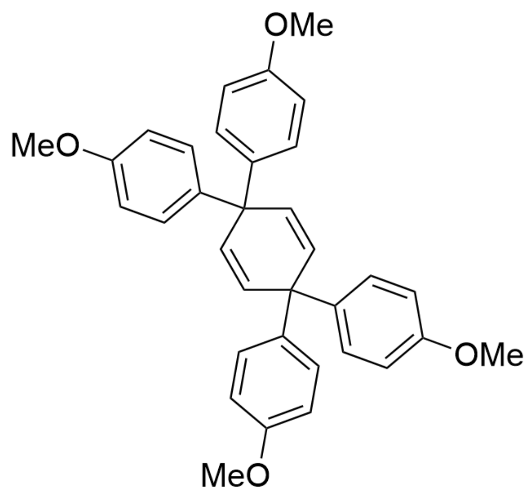
In carbon tetrachloride solution,



C1(C2(C3=CC=CC=C3)C=CC(C4=CC=CC=C4)(C5=CC=CC=C5)C=C2)=CC=CC=C1

reacts with $\mathrm{Br_2}$ at room temperature for 26h, yielding compound $\mathbf{A}(\mathrm{C_{30}H_{22}})$ with a yield of 98%.

If the reactant is changed to



COC1=CC=C(C2(C3=CC=C(OC)C=C3)C=CC(C4=CC=C(OC)C=C4)(C5=CC=C(OC)C=C5)C=C2)C=C1

, the resulting product is $\mathbf{B}(C_{34}H_{26}Br_4O_4)$.

There are the following statements about $\mathbf{A, B}$ and the reaction process:

1. The point groups to which the $\mathbf{A, B}$ molecules belong are the same when in the conformation with the highest symmetry.

2. If Br_2 is replaced with I_2, the point group to which the resulting product $\mathbf{B}'$ belongs in the conformation with the highest symmetry may be different from $\mathbf{B}$.

3. If the reaction time is further extended, the yields of $\mathbf{A, B}$ will remain unchanged.

4. The maximum number of rings in each intermediate that generates $\mathbf{A}$ is 5.

Then, the following options give all the correct options:

A. All other options are incorrect

B. 1

C. 2

D. 3

E. 4

F. 1, 2

G. 1, 3

H. 1, 4

I. 2, 3

J. 2, 4

K. 3, 4

L. 1, 2, 3

M. 1, 2, 4

N. 1, 3, 4

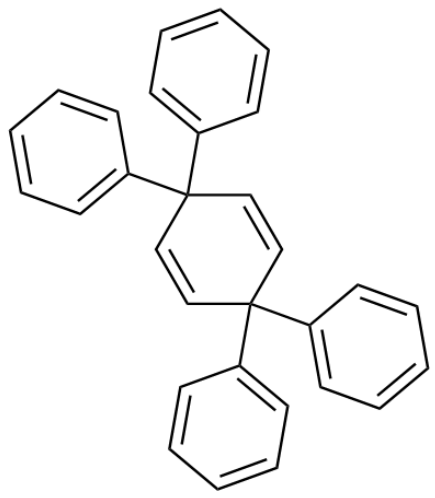
O. 2, 3, 4

P. 1, 2, 3, 4

Answer

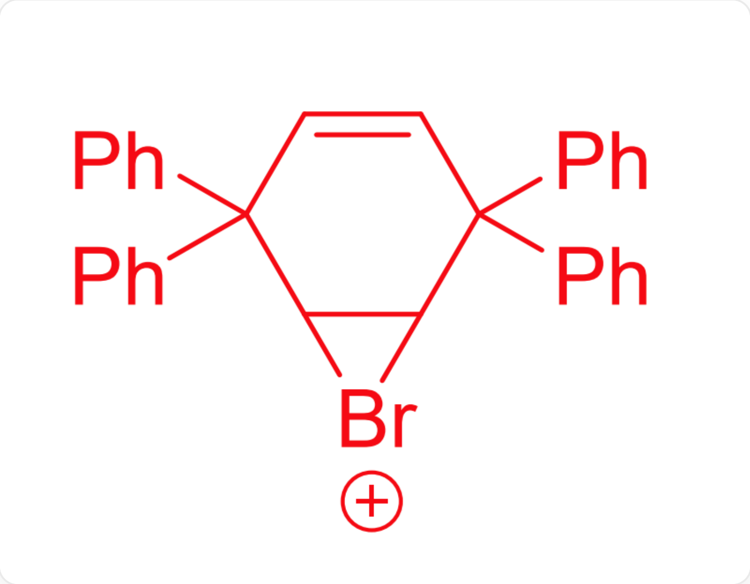
Correct Answer: F

Detailed Explanation



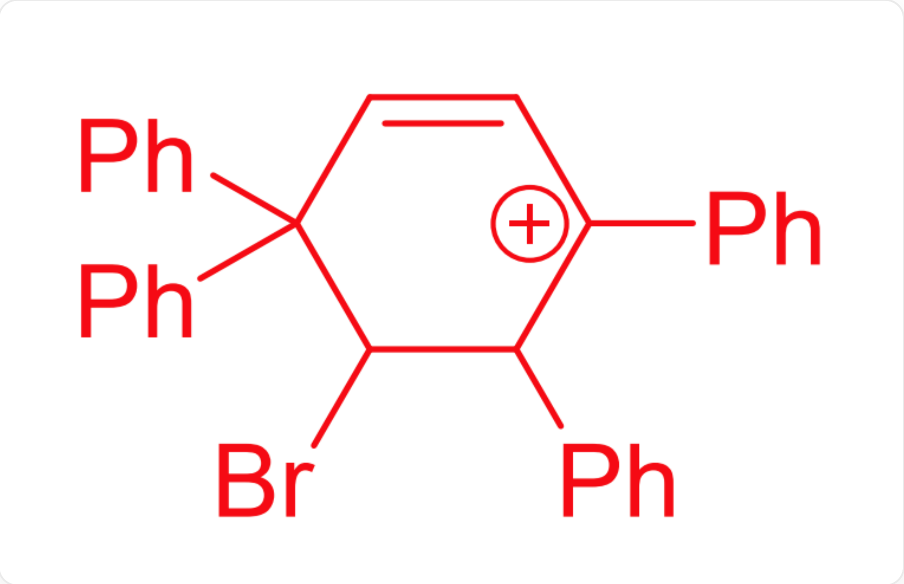
C1(C2(C3=CC=CC=C3)C=CC(C4=CC=CC=C4)(C5=CC=CC=C5)C=C2)=CC=CC=C1

Reacts with Br_2 to initially generate a bromonium ion intermediate:



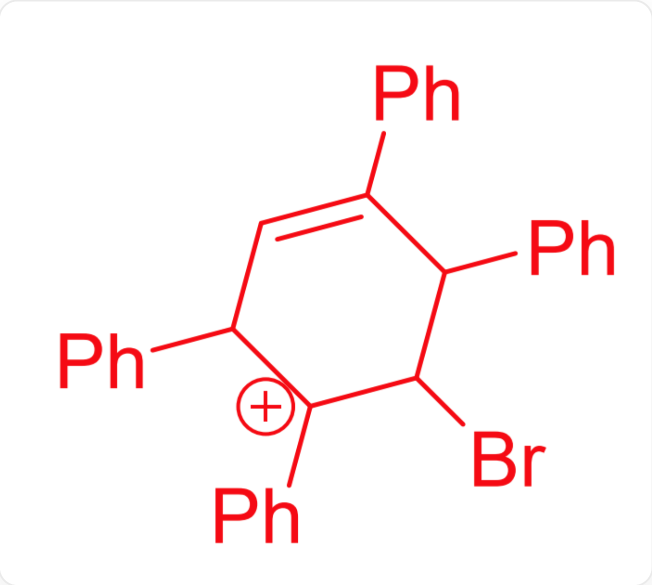
C12C(C3=CC=CC=C3)(C4=CC=CC=C4)C=CC(C5=CC=CC=C5)(C6=CC=CC=C6)C1[Br+]2

Subsequently, phenyl migration occurs, opening the three-membered ring to form a carbocation intermediate:



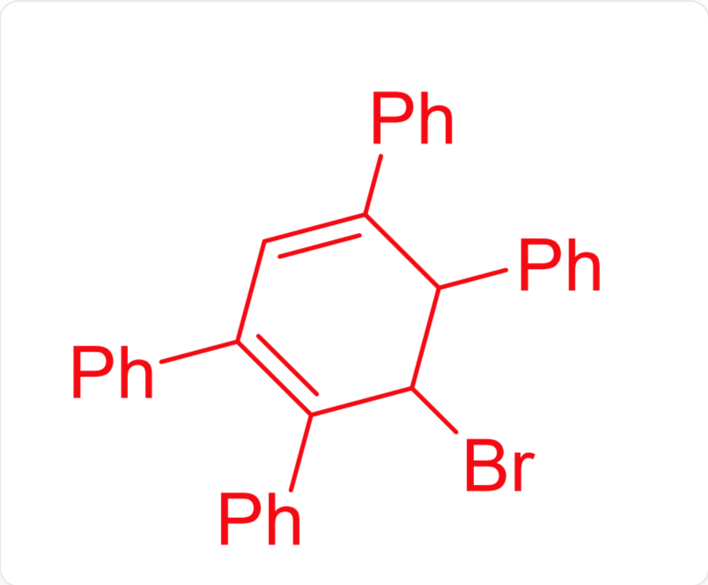
BrC1C(C2=CC=CC=C2)(C3=CC=CC=C3)C=C[C+](C4=CC=CC=C4)C1C5=CC=CC=C5

Then, another phenyl migration occurs, forming another carbocation intermediate:



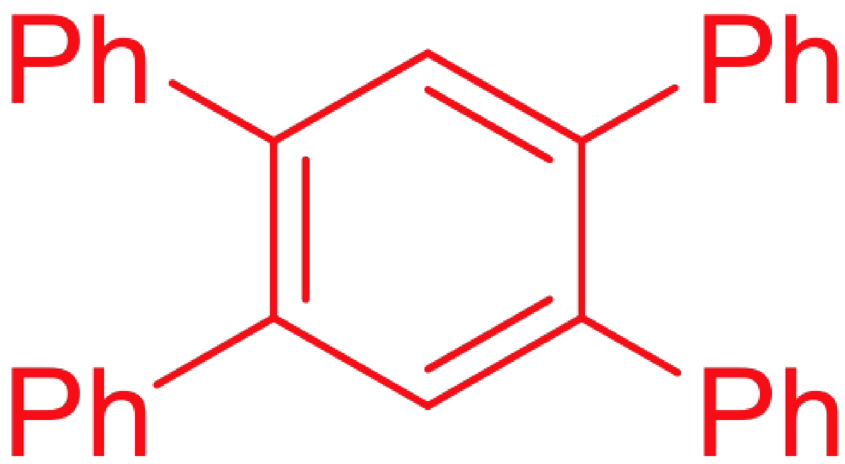
BrC1[C+](C2=CC=CC=C2)C(C3=CC=CC=C3)C=C(C4=CC=CC=C4)C1C5=CC=CC=C5

At this point, there are no groups available for migration, so the carbocation loses a proton to form a double bond:



BrC1C(C2=CC=CC=C2)=C(C3=CC=CC=C3)C=C(C4=CC=CC=C4)C1C5=CC=CC=C5

Finally, an elimination reaction occurs, forming a stable aromatic ring product:



C1(C2=CC=CC=C2)=C(C3=CC=CC=C3)C=C(C4=CC=CC=C4)C(C5=CC=CC=C5)=C1

CHECKPOINT 1 PTS

Important intermediate 1 for the reaction to generate **A**: C12C(C3=CC=CC=C3)(C4=CC=CC=C4)C=CC(C5=CC=CC=C5)(C6=CC=CC=C6)C1[Br+]2

CHECKPOINT 1 PTS

Important intermediate 2 for the reaction to generate **A**: BrC1C(C2=CC=CC=C2)(C3=CC=CC=C3)C=C[C+](C4=CC=CC=C4)C1C5=CC=CC=C5

CHECKPOINT 1 PTS

Important intermediate 3 for the reaction to generate $\mathbf{A}$: BrC1[C+](C2=CC=CC=C2)C(C3=CC=CC=C3)C=C(C4=CC=CC=C4)C1C5=CC=CC=C5

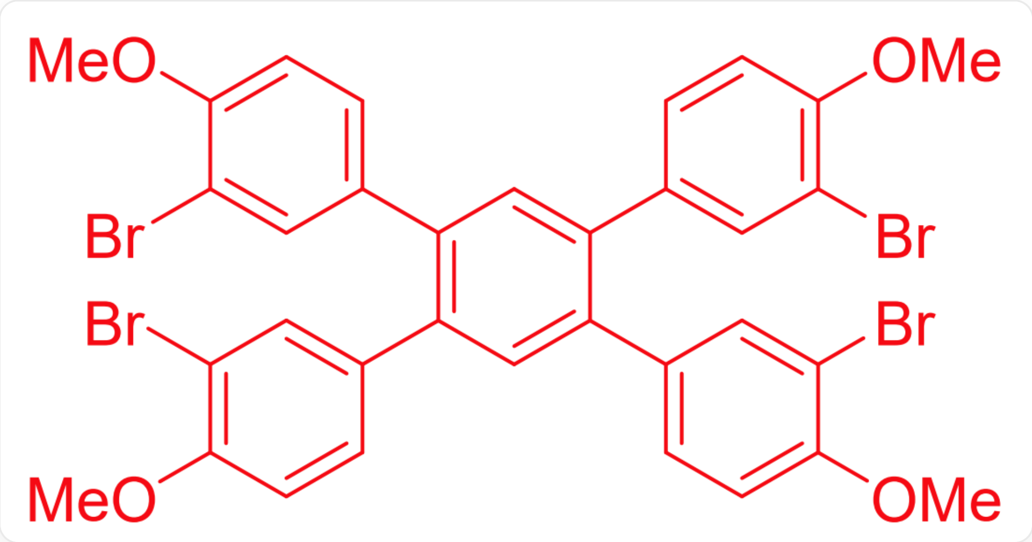
CHECKPOINT 1 PTS

Important intermediate 4 for the reaction to generate $\mathbf{A}$: BrC1C(C2=CC=CC=C2)=C(C3=CC=CC=C3)C=C(C4=CC=CC=C4)C1C5=CC=CC=C5

CHECKPOINT 1 PTS

$\mathbf{A}$ is C1(C2=CC=CC=C2)=C(C3=CC=CC=C3)C=C(C4=CC=CC=C4)C(C5=CC=CC=C5)=C1

The generation of $\mathbf{B}$ also involves the above process, first producing a product similar to $\mathbf{A}$. Noting that $\mathbf{B}(C_{34}H_{26}Br_4O_4)$ has four bromine atoms, it is speculated that electrophilic substitution has occurred on the benzene ring. Because of the presence of methoxy as an electron-donating group, $\mathbf{B}$ is:



BrC1=CC(C2=C(C=C(C(C3=CC(Br)=C(OC)C=C3)=C2)C4=CC=C(OC)C(Br)=C4)C5=CC(Br)=C(OC)C=C5)=CC=C1OC

CHECKPOINT 1 PTS

B is BrC1=CC(C2=C(C=C(C(C3=CC(Br)=C(OC)C=C3)=C2)C4=CC=C(OC)C(Br)=C4)C5=CC(Br)=C(OC)C=C5)=C(

The point group of molecules $\mathbf{A}, \mathbf{B}$ in the conformation with the highest symmetry are both D_{2h}, statement 1 is correct.

CHECKPOINT 1 PTS

The point group of molecules $\mathbf{A}, \mathbf{B}$ in the conformation with the highest symmetry are both D_{2h}

If Br_2 is replaced with I_2, the resulting product $\mathbf{B}'$ is structurally similar to $\mathbf{B}$, but due to the larger radius of the iodine atom, it may prevent the two benzene rings from being in the same plane, thereby disrupting the symmetry.

CHECKPOINT 1 PTS

The larger radius of the iodine atom may cause the product to lose its planar structure

If the reaction time is further extended, **A** is relatively stable and will not continue to react. **B** may continue to undergo electrophilic substitution reactions, thus changing the yield.

CHECKPOINT 1 PTS

B will continue to undergo electrophilic substitution reactions

It can be seen from the intermediates that the maximum number of rings in each intermediate that generates **A** is 6.

CHECKPOINT 1 PTS

C12C(C3=CC=CC=C3)(C4=CC=CC=C4)C=CC(C5=CC=CC=C5)(C6=CC=CC=C6)C1[Br+]2 has 6 rings

Therefore, statements 1 and 2 are correct.